

- 1 (a) Define *electric field strength*.

.....
..... [1]

- (b) Two flat parallel metal plates, each of length 12 cm, are separated by a distance of 1.5 cm, as shown in Fig. 1.1.

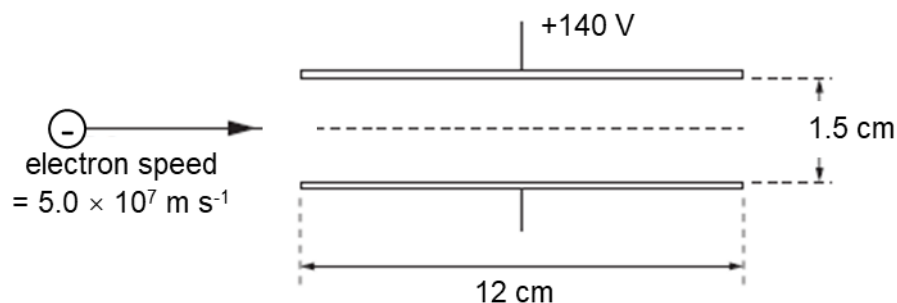


Fig. 1.1

The space between the plates is a vacuum. An upward electric field of field strength $1.40 \times 10^4 \text{ N C}^{-1}$ may be assumed to be uniform in the region between the plates and zero outside this region.

- (i) Given that the potential of the top plate is +140 V, determine the potential of the bottom plate.

potential of bottom plate = V [2]

- (ii) An electron initially travels parallel to the plates along a line mid-way between the plates, as shown in Fig. 1.1. The initial speed of the electron is $5.0 \times 10^7 \text{ m s}^{-1}$.

Assuming gravity and air resistance is negligible, for the electron between the plates,

1. determine the magnitude and direction of its acceleration,

acceleration = m s^{-2} [2]

direction = [1]

2. calculate the time for the electron to travel a horizontal distance equal to the length of the plates.

time = s [2]

3. Use your answers in **(b)(ii)1** and **(b)(ii)2** to determine whether the electron will hit one of the plates or emerge from between the plates.

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..... [3]

[Total: 11]